A Where the background comes from

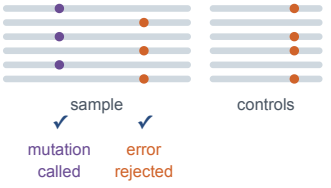
Read-level test (e.g. LoFreq, Geneious)



Each base is judged only by its own quality score. Sequencing errors that recur at one position (PCR, homopolymers, paralog reads) pass this test and are wrongly counted as SHM mutations, and reads from one animal are treated as independent replicates.

✓ correct decision ✕ wrong decision

Control-calibrated test (SHMtrace)



Error rates per substitution, strand and base quality are measured in matched AID-null libraries from the same run. A call needs support beyond this background (FDR ≤ 0.05), and groups are compared with one value per animal. This is the route for reads; a Sanger clone is counted instead (C).

B Inputs and read preparation



Nanopore
full-length amplicons



Sanger
.ab1 traces + sample sheet



Illumina
paired reads, merged



UMI-tagged
one consensus per molecule

Read preparation

1. **Length and expected-error gates**
   over-length reads and reads averaging > 0.01 errors per base
2. **Competitive alignment**
   minimap2 against target and paralog references; chimeras out
3. **Paralog-allele filter**
   reads with ≥ 10 paralog-specific bases removed
4. **Matched depth**
   random draw of the same number of usable reads per library

Sanger: traces are trimmed at quality ≥ 20 in place of step 1, and every clone is kept, so step 4 does not apply.

C Calling: a test for reads, a count for Sanger clones

Reads — nanopore, Illumina, UMI-tagged: one molecule read many times

1 Control error model
per substitution, strand and base quality; a margin chosen by leave-one-control-out

2 Test and FDR
support against expected errors; FDR ≤ 0.05 set from the controls; controls tested against each other

Sanger traces — one molecule read once

1 Count the mismatches
every substitution at base quality ≥ 20 is counted; one read of one molecule cannot be tested

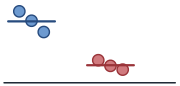
2 Background reported
AID-null clones give the calls the error alone would make, printed beside them and never removed

3 Germline and window
sites at ≥ 50% removed; calls restricted to the covered window; indels optional

4 Automatic checks
spectrum (C→T/G→A share), positional clustering, 5-mer targeting index, Sanger bookkeeping flags

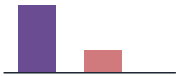
D Outputs

Animal-level tests and power



NB, Welch, permutation

Single-read estimate



observed minus control background

Clones from the V(D)J junction



reads grouped by junction; each clone counted once

Mutation trees



full-length reads of one clone

Repair-route indices



G:C Ts / Tv, A:T, WA/TW, decomposition

Reports, app and command line



HTML report, tables, project files